

**Sixth Semester B. Tech. (Electronics and Communication  
Engineering – Honors) Examination**

**MULTIMEDIA NETWORKS**

Time : 3 Hours ]

[ Max. Marks : 60

**Instructions to Candidates :—**

- (1) All questions carry marks as indicated.
- (2) Assume suitable data wherever necessary.

1. (a) Determine the VCI for incoming and outgoing packets for switches 1 and 2 in a virtual circuit network. Packets are routed in sequence through Source, switch 1, switch 2 and Destination. Consider the following information :

| Switch      | VCI |
|-------------|-----|
| Source      | 14  |
| Switch 1    | ?   |
| Switch 2    | ?   |
| Destination | 77  |

4(CO1)

- (b) Design a Banyan switch for 4 inputs and outputs. Also highlight the path for input 1 sending to output 0. 6(CO1)
2. (a) Considering the future multimedia networks enlist the dominant technologies in future. Justify the answer. 10(CO2)
3. (a) Compare ADPCM and PCM. 5(CO3)
- (b) Explain linear prediction algorithm. 5(CO3)

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|----|-----|------------------------------------------------------------|---------|
| 4. | (a) | Enlist the service improvements provided by H.323 network. | 3(CO3)  |
|    | (b) | Describe an H.323 terminal.                                | 7(CO3)  |
| 5. | (a) | Explain RTCP protocol.                                     | 10(CO3) |
| 6. | (a) | Explain multimedia call.                                   | 5(CO4)  |
|    | (b) | Elaborate 4G for multimedia communication.                 | 5(CO4)  |

